

WebCraft AI: Dynamic Web Prototype Synthesis and AI-Driven UI Mockup Generation

Tompea Maria-Daria
System Documentation & Future Roadmap

June 8, 2026

Abstract

This document outlines the architecture, design principles, and engineering specifications of **WebCraft AI**, an intelligent frontend interface that synthesizes user-selected keywords, typography configurations, and HSL palettes into personalized website mockups. The system leverages a dual-engine architecture: a frontend-driven Text-to-Image AI generation model (via Puter.js) for high-fidelity UI visual mockups, and a client-side layout compiler that renders fully interactive, responsive HTML/CSS code prototype structures inside a sandboxed viewport. Finally, we present a roadmap for next-generation expansion features, specifically focusing on how complex style combinations (e.g., *Artistic + Fitness*) can lead to highly custom, context-aware web layouts.

Contents

1	Project Overview	2
2	System Architecture	2
2.1	User Interface & State Machine	2
2.2	The Dual-Engine System	2
3	Design System: Palettes & Typography	2
3.1	Curated Palettes	2
3.2	Typography Framework	3
4	Dynamic Contextualization (e.g., Artistic + Fitness)	4
5	Future Roadmap: Mind-Blowing Features	4
5.1	Agentic Code Orchestration (LLM Multi-Agent System)	4
5.2	Neuro-Stylist (Visual-to-Code Pipeline)	4
5.3	Interactive 3D Environments & Physics	5
5.4	Emotional Semantic Styling	5

1 Project Overview

Traditional website generation platforms rely on static templates that require extensive manual modification. **WebCraft AI** shifts this paradigm by enabling users to describe their target website using a structured dictionary of 50 keywords divided into Industry, Vibe, Mood, Layout, and Target Audience categories. By combining these tag metrics with selected typography and color palettes, the system generates both a conceptual mockup image and a working prototype instantly, lowering the barrier to entry for modern web prototyping.

2 System Architecture

2.1 User Interface & State Machine

The application operates as a single-page application (SPA) running in the client browser, maintaining a central state object tracking:

1. **Selected Tags:** A bounded set of up to 8 tag parameters.
2. **Theme Parameters:** CSS color variables (Primary, Secondary, Canvas Background, Text) and dynamic Google Font references.
3. **Active Studio Context:** Switchboard for changing viewports (Desktop/Tablet/Mobile) and tabs (Mockup/Prototype/Source Code).

2.2 The Dual-Engine System

The core logic resides in a bifurcated generation pipeline:

- **Image Engine (Puter.js AI):** Integrates with the browser-based Puter.js AI SDK to execute a text-to-image API request asynchronously. The prompt is automatically structured from the active tags, typography, and color tokens, compiling it into an optimized Stable Diffusion description.
- **Prototype Engine (Local Compiler):** Dynamically compiles raw HTML, custom CSS rules, and semantic scripts. It formats the page layout, imports Google Fonts, overrides CSS custom properties with selected HSL/Hex color values, and writes the output directly to a sandboxed iframe using ‘srcdoc’ streaming.

3 Design System: Palettes & Typography

3.1 Curated Palettes

WebCraft AI maps color combinations to HSL color spaces to ensure readable contrast ratios. Curated default theme presets include:

Table 1: Curated Theme Palette Configurations

Preset Name	Primary Color	Secondary Color	Canvas Color	Text Color
Midnight Neon	#8B5CF6	#06B6D4	#09080E	#F3F4F6
Sunset Minimalist	#F97316	#E11D48	#FAFAF9	#1C1917
Nordic Spruce	#10B981	#3B82F6	#0F172A	#F8FAFC
Brutalist Peach	#FF3E3E	#FACC15	#FFFBEB	#1C1917
Cyberpunk Grid	#FF007F	#00F0FF	#050508	#FFFFFF

3.2 Typography Framework

The typography engine imports Google Fonts dynamically at runtime. It loads serif styles for traditional, organic, or elegant vibed websites, monospace fonts for developer/hacker portfolios, and modern geometric sans-serif fonts for SaaS dashboards. Examples of loaded families include *Outfit*, *Plus Jakarta Sans*, *Space Grotesk*, *Syne*, *Cabinet Grotesk*, and *JetBrains Mono*.

4 Dynamic Contextualization (e.g., Artistic + Fitness)

To solve the issue where selecting different tag pairs (such as **Artistic** + **Fitness**) feels too generic, we propose a rules-based layout engine that adapts components based on selected tag combinations:

Traditional Fitness Site	Artistic + Fitness Site
• Rigid grid alignment • Solid backgrounds • Standard list items (Classes, Trainer) • Corporate blue/red accents	• Overlapping asymmetrical sections • Liquid glassmorphism, paint brush highlights • Dynamic workout movement gallery • Neon organic gradients, custom fonts

When **Fitness** is selected alongside **Artistic** or **Retro**, the layout compiler shifts:

- **Structural Shifting:** Switches from a structured corporate grid to an overlapping, asymmetrical layout with floating canvas elements.
- **UI Element Treatment:** Converts standard card grids into glassmorphic cards with custom paint-like border highlights and curved organic borders ('border-radius: 40px 10px 40px 10px').
- **Content Copy:** Overwrites generic fitness text with creative slogans, e.g., "Sculpt Your Canvas: Where Physiology Meets Fine Art."

5 Future Roadmap: Mind-Blowing Features

5.1 Agentic Code Orchestration (LLM Multi-Agent System)

Instead of relying on rigid, pre-built template compilers, future generations will delegate generation to a team of micro-agents:

- **Layout Architect Agent:** Evaluates user keywords and designs a custom JSON representation of the layout tree.
- **Tailwind Design Agent:** Generates unique components and styles them with customized utility classes matching the vibe.
- **Interactive Animation Agent:** Injects custom micro-animations (e.g., Framer Motion or GSAP scroll-triggered actions) matching the vibe.

5.2 Neuro-Stylist (Visual-to-Code Pipeline)

Currently, the AI mockup image and the interactive code prototype are generated separately. The future architecture will link these using a Visual-to-Code pipeline:

1. Generate the UI mockup image via Stable Diffusion / Midjourney.
2. Feed the generated image into a multimodal Vision model (like Gemini 1.5 Pro).
3. The model extracts CSS variables, element coordinates, typography spacings, and layout flow to write HTML/Tailwind code that matches the visual layout pixel-for-pixel.

5.3 Interactive 3D Environments & Physics

Websites under the *Gaming*, *Futuristic*, or *Artistic* tags will dynamically load interactive 3D assets:

- Dynamically load a Canvas and initialize a lightweight 3D web viewer (using Three.js / Spline).
- Embed interactive particles that respond to mouse movements, or integrate a simple physics engine (like Matter.js) where website layout cards fall and react to gravity inside the viewport.

5.4 Emotional Semantic Styling

Instead of relying on manual color pickers, the system will run sentiment analysis on selected tags. Selecting tags like *Vibrant + Playful + Kids* will generate pastel-toned bouncing bouncy physics shapes, whereas tags like *Luxury + Elegant + Dark* will auto-generate gold-leaf borders, soft radial gradients, and slow cinematic reveal transitions on scroll.